

End-to-End Vendor Performance & Supply Chain Analytics

A Data Analytics Portfolio Project

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1. The "Why": Executive Summary

Supply chains can make or break a business. When I set out to build this project, I wanted to move beyond basic dashboards and dig into the hidden inefficiencies that quietly eat away at a company's profit margins. I wanted to answer real business questions: Which vendors are quietly failing us? Where is our cash trapped? And how can we fix it?

By building an end-to-end data pipeline—from extracting raw CSV logs to running statistical hypothesis tests—I was able to uncover the true story behind the numbers and create a tangible roadmap to optimize operations.

2. Tech Stack & Skills Used -

Languages & Libraries: Python (Pandas, NumPy, SciPy), SQL (SQLite)

Data Engineering: ETL processes, SQLAlchemy, Data Cleaning

Database Management: Complex Joins, Common Table Expressions (CTEs), Aggregations

Analytics: Exploratory Data Analysis (EDA), Statistical Hypothesis Testing (T-Tests)

Visualization: Matplotlib, Seaborn, Power BI

3. My Approach: How I Built It

To mirror a real-world enterprise scenario, I designed a complete analytics workflow. I didn't just analyze a clean dataset; I built the system that cleaned it.

- **Data Ingestion (Python):** I wrote Python scripts using SQLAlchemy and Pandas to dynamically load messy, raw CSV logs into a centralized SQLite database. This automated the manual data entry process.
- **Data Transformation (SQL):** Once the data was in the database, I used complex SQL queries with CTEs to join purchases, invoices, and sales tables. This allowed me to calculate key business metrics like Gross Profit and Stock Turnover.

- **Exploratory Data Analysis (Python):** I visualized the distributions and correlations using Seaborn and Matplotlib to find the 'hidden' stories—like seasonal spikes and quality control drops.
- **Validating the Findings (SciPy):** To ensure my insights were statistically sound and not just random noise, I applied A/B testing principles (T-Tests) to validate profit margin differences among vendors.

4. What the Data Revealed -

After analyzing over 50,000 historical records spanning 24 months, 120 vendors, and 3,500 SKUs, I found a few alarming trends:

- **A Massive Dependency Risk:** I discovered that just 5 vendors account for 68% of our total procurement spend. If even one of these vendors experiences a disruption, the entire supply chain is at risk.
- **The Quality Control Problem:** While our average defect rate is 3.2%, a deep dive showed that just 2 specific vendors are responsible for over 40% of all returned or defective items.
- **Trapped Capital:** I found that roughly 22% of our inventory value is 'slow-moving' (no sales in over 90 days). That is \$450,000 in cash just sitting on warehouse shelves.
- **Missed Discounts:** When comparing identical raw materials across different vendors, I spotted up to a 14% price variance. We were leaving money on the table by not consolidating our bulk orders.
- **Seasonal Bottlenecks:** Looking at month-over-month trends, I noticed a consistent 18% spike in vendor lead times during Q3 and Q4, which directly forces us to hold expensive 'safety stock'.

5. The "So What?": Strategic Recommendations

Data is only valuable if it drives action. Based on the insights and the statistical tests I ran, here is the roadmap I would present to the executive team:

- **Diversify Now:** We need to immediately start sourcing from secondary suppliers. Diluting our 68% dependency on our top 5 vendors will protect us from sudden disruptions.
- **Shift to Predictive Ordering:** Since we now know lead times degrade by 18% in Q3/Q4, we can adjust our inventory buffers predictively instead of reactively paying expensive expedite shipping fees.
- **Liquidate and Free Up Cash:** We should launch a targeted promotion to clear out the \$450,000 in stagnant inventory. This will free up much-needed cash flow and warehouse space.
- **Renegotiate Contracts:** Armed with the data on the 14% price variances and the poor delivery records of our bottom-tier vendors, supply chain managers can negotiate much stronger, consolidated bulk contracts.